

Lecture 5: Evaluation & Experiment Design

41-501
Applied AI

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Content

- Metrics
- Data & Baselines
- A/B Testing
- Experiment Tracking
- Summary (+Lab)



The Accuracy Trap

- Fraud detection: 99.7% of transactions are legitimate
- A model that always predicts "not fraud" → 99.7% accuracy

→ Useless



Classification Metrics

- Precision: "Of everything flagged, how much was actually positive?" → How much do I trust an alarm?
- E.g. spam filter
- Recall: "Of all actual positives, how many did I catch?" → How many did I miss?
- E.g. Airport Security
- F1: Harmonic mean: balances both

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

		Predicted 0	Predicted 1
Actual	0	TN	FP
	1	FN	TP

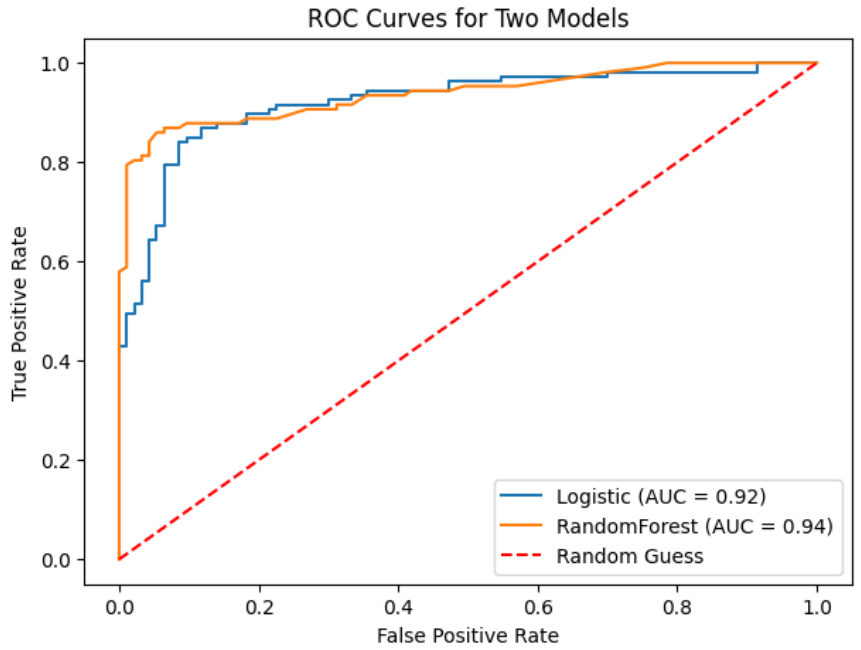
$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$



AUC-ROC

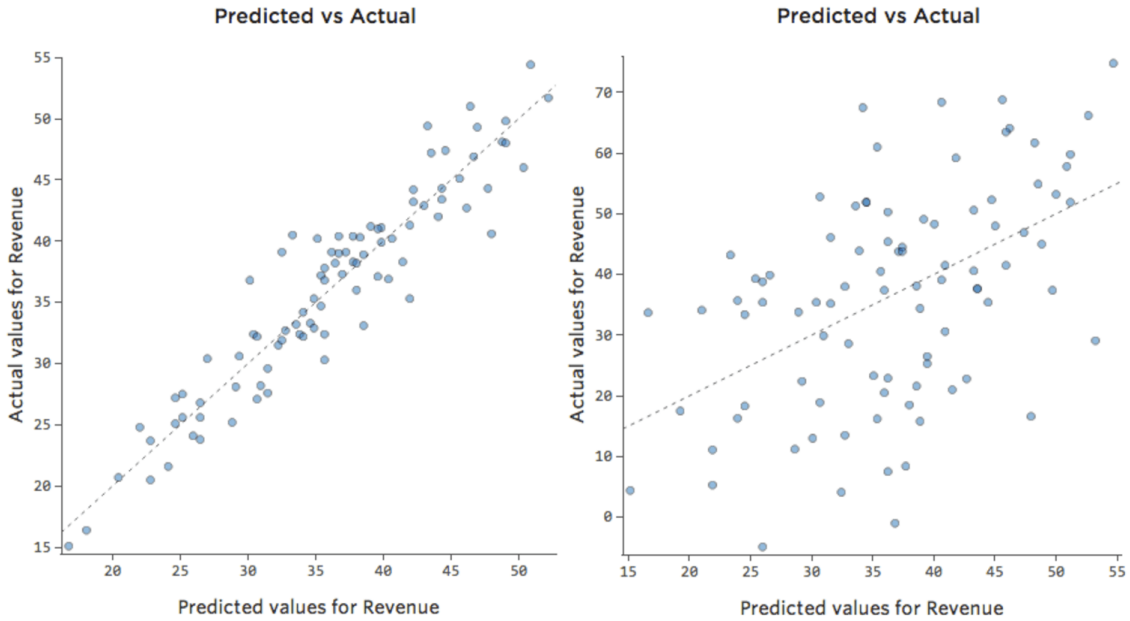
- ROC (receiver operating characteristic) curve
  True positive rate vs false positive rate at every threshold
- AUC (area under the curve)
  - 0.5 = random
  - 1.0 = perfect.
- Useful when you'll tune the decision threshold later.





Regression Metrics

- MAE (mean absolute error): robust to outliers, interpretable ("off by €X on average")
- RMSE (root mean square error): penalizes large errors more heavily
- R^2: proportion of variance explained (good for communication, bad alone)



Ranking Metrics

- When order matters, not just right/wrong → e.g. recommendations, search results
- MRR (Mean Reciprocal Rank)
  - “Is the best item on top?”
  - → single relevant item
- MAP (Mean Average Precision)
  - "On average, how good is the ranking at each relevant item?”
  - → multiple relevant items
- NDCG (Normalized Discounted Cumulative Gain)
  - "Are the best items near the top?”
  - → order is important



The Metric Is a Product Decision

- Startup: "20% lift in conversion rate"
  → conversion rate is your metric
- Research: "Advance SOTA on benchmark X"
  → benchmark's metric
- Industry: "Cut false alerts by 50%, keep catch rate above 95%"
  → precision at recall ≥ 0.95
- Metric is not a technical choice (→ stakeholder interest)



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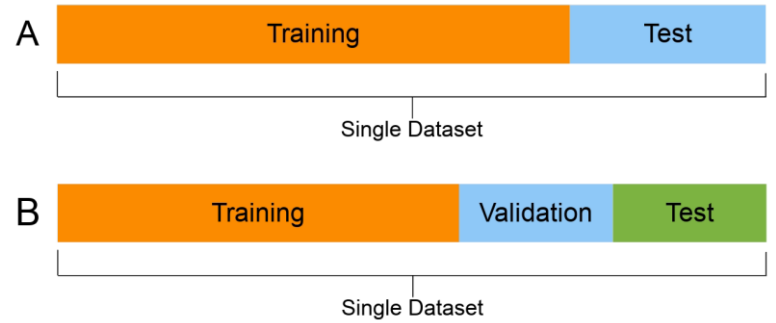


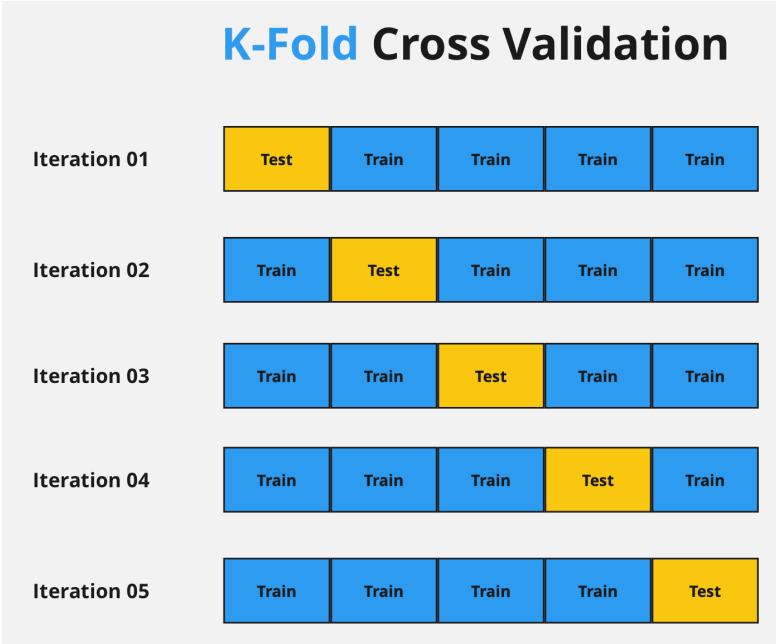
Train / Validation / Test

- Train: learn patterns
- Validation: tune hyperparameters
- Test: final honest evaluation

- Why not just train/test? Tuning on the test set → indirectly fitting to it
- Cross-Validation: K-Fold: split data into k parts, train on k-1, validate on 1, rotate (especially important with small datasets)

- Golden rule: test set is touched once, at the end!





Data Leakage (Again!)

- Time-Series Split: never shuffle temporal data. Train on past, validate on future.
- Information from the future or test set leaks into training.
- Your model looks amazing in development. Falls apart in production.
- "If your results look too good to be true, they probably are."



Common Leakage Sources

- Fitting a scaler/encoder on the full dataset before splitting
- Solution: Use sklearn Pipeline: scaler + model in one object, fit only on training data.
- Features derived from future timestamps (e.g. "total purchases this month" available on day 1)
- Solution: Think carefully about features: "Would I have this information at prediction time?”
- Duplicate samples landing in both train and test
- Solution: Check for duplicates across splits.

- Be suspicious of perfect or near-perfect results.

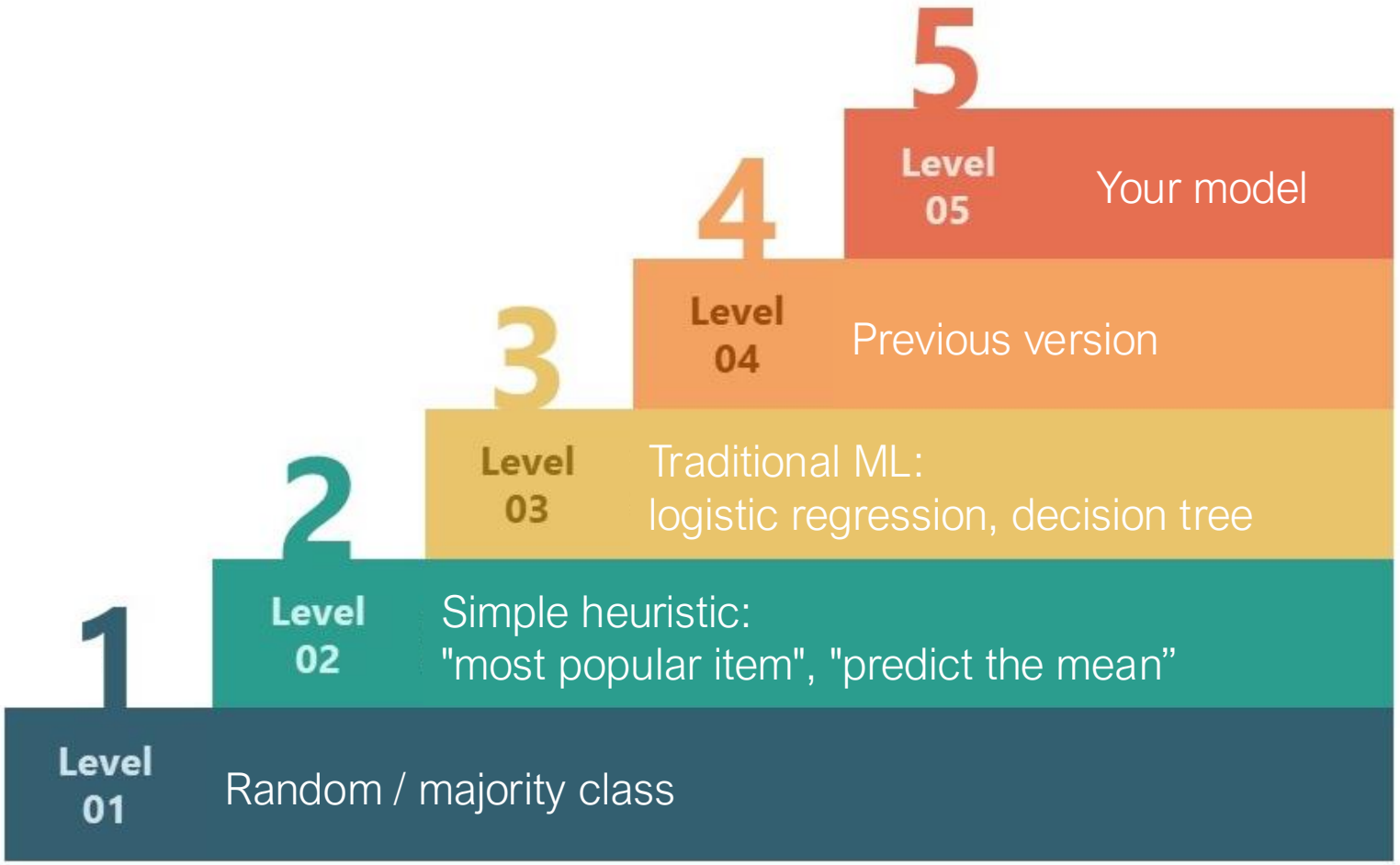


Why Baselines Matter

- "Our model achieves 87% F1.”: Is that good?
- Without a baseline, you can't tell. 87% could be worse than a coin flip on easy data, or remarkable on hard data.
- Always start with the simplest thing that could work.



Types of Baselines

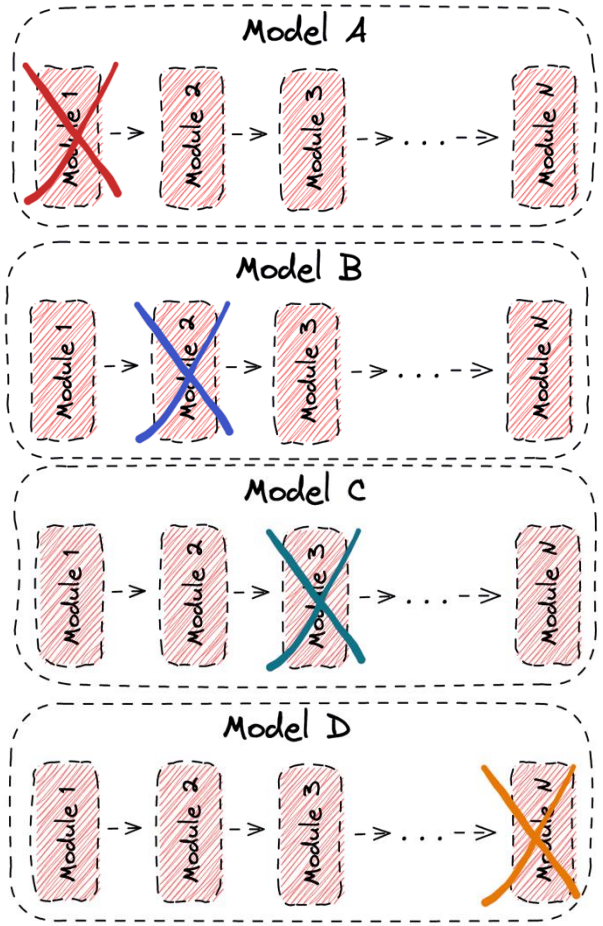


Ablation Studies

- Remove one component at a time to see what actually matters.
- Example:
  - Full model 91%
  - Without attention: 88%
  - Without pretraining: 84%
  - Without augmentation: 90%.

→ Pretraining contributes the most. Augmentation barely helps.

- The research equivalent of A/B testing.



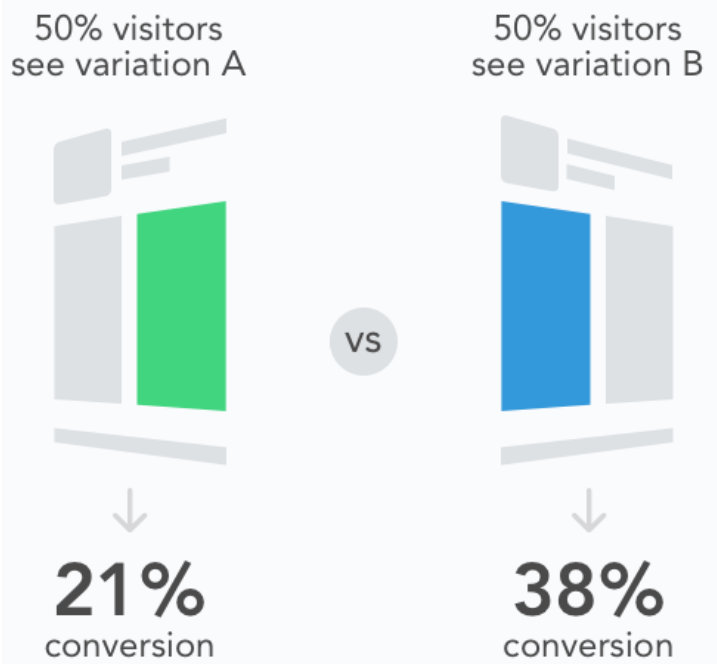
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A/B Testing: The Gold Standard

- The question: "Does this change actually help real users?”
- Split users randomly into two groups. A gets the current system, B gets the new one. Measure and compare.
- Used by every major tech company: Netflix, Booking.com, Google, …





The A/B Testing Process

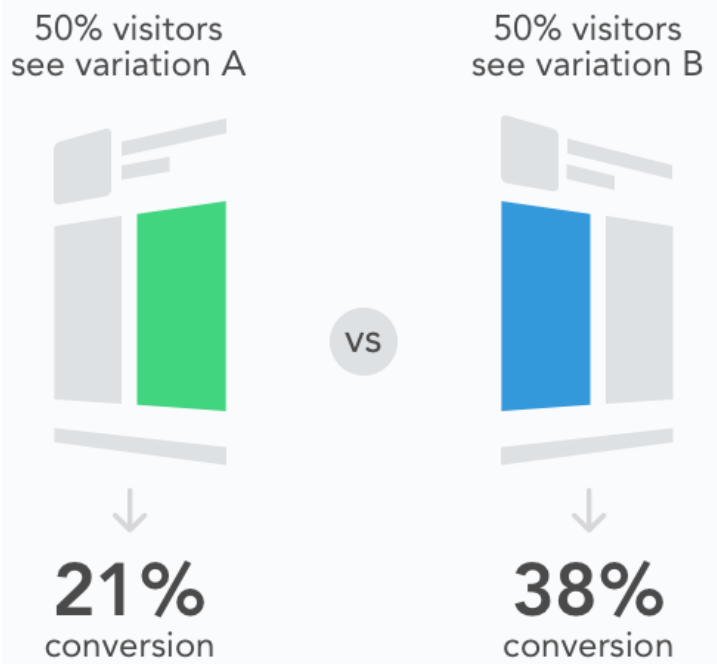
1. Define hypothesis: "Model B increases conversion rate”
2. Choose metric: conversion rate (purchases / visits)
3. Determine sample size (how many users you need → before starting!)
4. Run the experiment (randomly assign users)
5. Analyze results
6. Decide: ship, iterate, or kill





Example: E-Commerce Recommendations

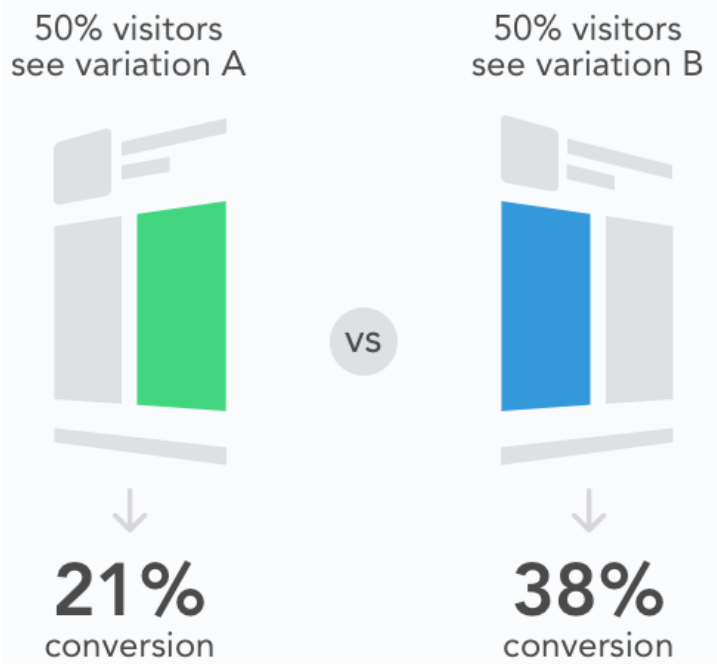
- Current system (A): popularity-based, 3.2% conversion rate
- New ML model (B): collaborative filtering, ??? conversion rate
- 5,000 users per group, 2 weeks.
- Result: A converts at 3.1%, B converts at 3.7%.
- Is B actually better, or was this just luck?





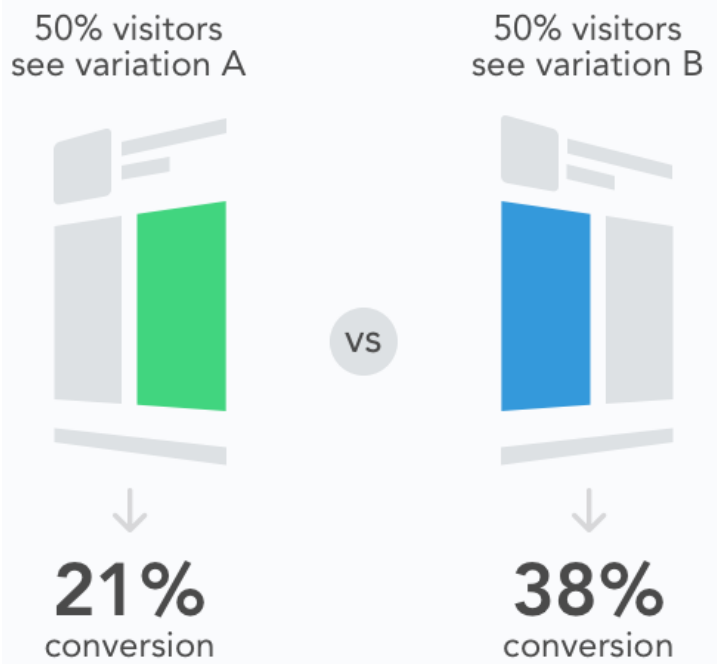
Statistical Significance

- p-value: probability of seeing this difference (or more extreme) if there were actually no difference.
- Large p-value: probably random variation
- Small p-value: unlike just chance
- Convention: $p < 0.05 \rightarrow$ "statistically significant"
- Means: "unlikely to be pure chance."



A/B Testing Pitfalls

- Peeking: checking results daily and stopping when you see $p < 0.05$ → inflated false positives
- Small samples: underpowered tests can't detect real effects
- Multiple comparisons: testing 20 metrics → one will be "significant" by chance
- Selection bias: groups must be truly random



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Experiment Tracking: The Problem

"Which hyperparameters gave me that good result last Tuesday?"

"I know I tried that combination already, but I can't find it."

"Wait, did I use the cleaned dataset or the raw one for that run?"



Experiment Tracking

Run	Model	LR	Depth	F1	AUC	Notes
001	RF	/	10	0.83	0.91	baseline
002	GB	0.1	5	0.86	0.93	better

- Log every experiment:
  - Parameters
  - Metrics
  - data version
  - code version
  - timestamp
  - …
- Even a spreadsheet is better than nothing.



Tools: MLflow + Weights & Biases

- MLflow: open-source, self-hosted, widely used in industry. Logs params, metrics, models.
- W&B: cloud-based, great visualizations, popular in research. Automatic dashboards.
- Both free for personal / academic use.







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Summary: Minimum Viable Evaluation

Always check at minimum:

- ✓ A clearly justified metric: not just accuracy
- ✓ A baseline comparison: prove your model adds value
- ✓ Proper evaluation: no leakage, honest train/test split
- ✓ Confidence in your results: error bars, cross-val, or significance test



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Thank you!

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